Orbit-Band Projection and Mesoscopic Spectral Transfer for an Actual-Coefficient Determinant Sector

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Abstract

The large-prime determinant sector in the twin-prime-correlation (TPC) sequence has a smooth orbit variable but an output vector whose orbit coordinate is not uniformly band limited. This is the first of four transfer gates left by the preceding shift-spectral analysis. We act on the literal residual output coordinate: after zero extension in the physical orbit variable, let Π_Ω be the fiberwise orthogonal Fourier projection onto the half-width-Ω comb around $H_0^{-1}\mathbb{Z}/\mathbb{Z}$, leaving the opposite row and the physical residual label $s = d(m)r$ unchanged. For a progression-restricted orbit profile with Fourier bandwidth Σ, we prove that the projected completed determinant packet has the exact shift-spectral gap

$$\frac{\Omega + \Sigma/J}{M_-} < \|\alpha\|_{\mathbb{T}} < \frac{H_0^{-1} - \Omega - \Sigma/J}{M_+}.$$

The theorem is Hilbert valued and permits arbitrary arithmetic coefficients on the prime, cofactor, and source variables. We then apply a canonical comb cutoff to the complete vector-valued actual TPC source–mask profile; the proved projective representation is used only to estimate its complement. The discarded tails have super-polynomially small leakage, so the projected frozen shift family built from the literal physical h_0-coefficient field inherits rapid decay throughout a slightly shrunken gap. At the inherited TPC endpoint, an orbit band with time–bandwidth size $X^{1/400+o(1)}$ retains the nonempty range

$$X^{-399/400+o(1)} < \|\alpha\|_{\mathbb{T}} < X^{-267/400+o(1)}.$$

The projection is contractive, commutes with the residual Möbius unitary, and has exact physical-window trace $2H_0N\Omega$ per scalar non-orbit fiber. It is injective on every nonzero finitely supported orbit vector; however, an explicit finite-difference family shows that no uniform fraction of the residual norm must lie in this band. Thus an actual-residual-output sector with growing time–bandwidth trace crosses the orbit-frequency gate, while the high-frequency complement, atomic normalization, alias unfolding, and fixed-shift localization remain open. At the physical intercept this is the actual-coefficient residual synthesis, not the native affine physical Gram face. The shift family outside that intercept is frozen-coefficient, and no fixed-shift TPC closure, parity breach, or twin-prime conclusion is claimed.

Keywords: determinant equation; orbit-frequency projection; shift spectrum; Möbius correlation; Poisson summation; time–bandwidth concentration; parity problem.

Convention. We write $\mathrm{e}(x) = \exp(2\pi i x)$, identify $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and put $\|\alpha\|_{\mathbb{T}} = \min_{k\in\mathbb{Z}} |\alpha - k|$. Fourier transforms on $\mathbb{R}$ use $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)\mathrm{e}(-x\xi)\,dx$; the unitary Fourier transform on $\ell^2(\mathbb{Z})$ is $(\mathscr{F}y)(\beta) = \sum_{j\in\mathbb{Z}} y_j \mathrm{e}(-j\beta)$.

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1 Introduction

The preceding large-prime descent turns one part of the TPC equality column into the determinant equation

$$pr - mj = h_0, \tag{1}$$

with the prime variable p, a Möbius-weighted cofactor r, and an actual source row m. Projective reassembly separates the physical source–mask coefficient into those arithmetic variables and a smooth orbit profile, up to a fixed progression in j [2, 4]. TPC–46 then proves an exact mesoscopic shift gap when the Hilbert vector attached to (p, r, m) is independent of j [5]. In the actual residual output, the literal label $s = d(m)r$ remains fixed for fixed (m, r), but the non-orbit coefficient vector over opposite rows varies with j; its coordinates are

$$(\text{opposite row}, j, s), \qquad s = d(m)r. \tag{2}$$

TPC–47 records that dependence by adjoining the literal e_j factor and projecting it. Before projection, an arbitrary dual orbit field can restore the entire spectrum.

This paper does not assume that this coupling is smooth. It applies a bounded projection to the coupling itself. The physical orbit coordinate is embedded by zero extension into $\ell^2(\mathbb{Z})$. In every fixed opposite-row and residual-label fiber, we retain the full residue-comb Fourier band

$$\mathcal{B}_\Omega^{(H_0)} := \{\beta \in \mathbb{T} : \operatorname{dist}(\beta, H_0^{-1}\mathbb{Z}) \le \Omega\}. \tag{3}$$

This defines one canonical sector of the actual output Hilbert space; it is not the choice of one term in a signed projective expansion. The source mask, the Möbius cofactor, and the residual label remain literal.

1.1 The positive theorem

The abstract kernel underlying the result is transparent. Let $M_- \le m \le M_+$, let H_0 be fixed, and consider one layer

$$F_h = \sum_{p,r,m} \sum_{\substack{j \in \mathbb{Z} \\ j \equiv a \pmod{H_0}}} A_{p,r,m} \phi(j/J) v_{p,r,m} \otimes e_j \mathbf{1}_{pr-mj=h}, \tag{4}$$

where $v_{p,r,m}$ is independent of j. If $\operatorname{supp} \widehat{\phi} \subset [-\Sigma, \Sigma]$, then the orbit Fourier variable β changes the Poisson kernel from $m\alpha$ to $m\alpha - \beta$. The exact nonresonance interval is therefore

$$\boxed{\frac{\Omega + \Sigma/J}{M_-} < \|\alpha\|_{\mathbb{T}} < \frac{H_0^{-1} - \Omega - \Sigma/J}{M_+}}. \tag{5}$$

On this interval the shift Fourier transform of $\Pi_\Omega F$ vanishes identically. The statement is vector valued, uses no cancellation in p, r, m, and is uniform over their coefficients.

The actual orbit profiles are smooth rather than band limited. We use a fixed smooth cutoff in continuous Fourier space,

$$\phi = \phi_{\le \Sigma} + \phi_{> \Sigma}, \qquad \operatorname{supp} \widehat{\phi_{\le \Sigma}} \subset [-\Sigma, \Sigma]. \tag{6}$$

Every Schwartz seminorm of the remainder is smaller than an arbitrary negative power of Σ, provided a sufficiently high but fixed orbit seminorm of ϕ is used. TPC–36 controls the total signed projective mass with every fixed orbit seminorm. Hence the tails in (6) can be summed over the *entire* actual projective representation. No layer is selected and no arithmetic coefficient is smoothed.

At the endpoint

$$M_\pm = X^{267/400+o(1)}, \qquad J = X^{133/400+o(1)}, \tag{7}$$

take

$$\Omega = X^{1/400+o(1)}/J, \qquad \Sigma = X^{1/400+o(1)}. \tag{8}$$

After a harmless X^ϵ shrink at both ends, (5) becomes

$$\boxed{X^{-399/400+\epsilon} \le \|\alpha\|_{\mathbb{T}} \le X^{-267/400-\epsilon}.} \tag{9}$$

The projected frozen shift family built from the literal physical h_0-coefficient field has leakage $O_{A,\epsilon}(X^{-A})$ on this interval relative to any of the declared polynomial coefficient envelopes. Thus the orbit-frequency gate is crossed on a growing sector whose time–bandwidth size is the same $X^{1/400+o(1)}$ scale as the endpoint TPC allowance.

1.2 What the projection does and does not retain

On one fixed opposite-row and residual-label fiber, let E_I restrict the orbit coordinate to a physical interval of $N \asymp J$ integers. The band projection satisfies

$$\|\Pi_\Omega\| = 1, \qquad \|\Pi_\Omega E_I\|_{\mathrm{HS}}^2 = \mathrm{tr}(E_I^* \Pi_\Omega E_I) = 2H_0 N\Omega. \tag{10}$$

Thus $2H_0N\Omega \asymp X^{1/400+o(1)}$ is an exact trace identity, not an informal count of Fourier modes. Moreover, a nonzero finite-orbit vector cannot have zero projection onto an open frequency band: its Fourier transform is a nonzero trigonometric polynomial. The projected sector is therefore genuinely nonempty whenever the physical residual packet is nonzero.

There is nevertheless no useful uniform lower bound for the retained mass. Explicit finite-difference vectors y_n, supported inside a physical orbit window, satisfy

$$\frac{\|\Pi_\Omega y_n\|_2}{\|y_n\|_2} \ll n^{1/4}\Omega^{1/2}(C_{H_0}\Omega)^n. \tag{11}$$

Consequently, the theorem controls a canonical actual-residual-output sector but does not control the orthogonal complement or the norm of the full equality column. This is a sharp distinction between a spectral transfer theorem and a fixed-shift TPC estimate.

1.3 Claim ledger

The results are organized as follows.

1. Section 2 identifies the physical large-prime packet and its frozen-coefficient shift extension.
2. Section 3 constructs the literal fiberwise orbit projector, proves its exact trace, and records the commuting residual structure.
3. Section 4 proves (5), including the fixed progression.
4. Section 5 gives the low-pass/tail decomposition and quantitative leakage bounds.
5. Section 6 transfers those bounds to the full actual source–mask coefficient packet.
6. Section 7 proves the rational endpoint ledger.
7. Section 8 proves nonvanishing but disproves a uniform retained-mass fraction.
8. Section 9 separates the remaining gates, and section 10 records the exact finite certificate and claim boundary.

Only the column $h = h_0$ in the frozen family is the inherited physical equality column. The other shifts keep its coefficient field fixed in order to expose the spectrum. We do not identify them with TPC packets reconstructed at new physical shifts. We also do not deduce atomic normalization, alias unfolding, a deterministic fixed-shift Möbius estimate, a parity breach, or a prime-pair asymptotic.

2 The inherited physical-coefficient sector

We record the exact object to which the orbit projection is applied. The purpose is to prevent a low-frequency statement about a model coefficient from being substituted for the actual source–mask field.

2.1 The physical equality column

Fix $h_0 \in \mathbb{Z} \setminus \{0\}$, and put $H_0 = |h_0|$. A source row is $\rho = (\ell_\rho, d_\rho)$, with

$$m_\rho = \ell_\rho d_\rho \asymp Q, \qquad Q = X^{267/400+o(1)}, \qquad J = X^{133/400+o(1)}. \tag{12}$$

Let M_- and M_+ be the minimum and maximum source integers on the inherited dyadic row cell. Thus

$$0 < M_- \le m_\rho \le M_+, \qquad M_- = X^{267/400+o(1)}, \qquad M_+ = X^{267/400+o(1)}. \tag{13}$$

The prime ℓ_ρ and squarefree d_ρ are uniquely determined by m_ρ on the inherited row support. Let $\mathcal{O}_L$ and $\mathcal{O}_R$ be the opposite-row sets for the left and right columns. We treat the left column; every assertion has the symmetric right-column form.

On one terminal block $n \asymp X$, take the large-prime band

$$\mathcal{P}^\sharp_{R_0} = \left\{ p \in \mathbb{P} : \sqrt{N_+} < p \le N_-/R_0 \right\}. \tag{14}$$

Writing $n = pr$, TPC–45 proves that p is unique, that $R_0 \le r < \sqrt{N_+}$, and that the physical residual label is

$$s = d_\rho r. \tag{15}$$

After zero atoms are removed, d_ρ, r, s are squarefree on the equality column and $\mu(s) = \mu(d_\rho)\mu(r)$ [3–5].

Let $\gamma^{(1)}_\rho$ be the physical source coefficient, $\mathfrak{A}^{\mathrm{act}}_{\rho,\gamma}(j)$ the actual joint source–mask multiplier, and

$$\Lambda(n) = (\log n)\Phi_0(n/X). \tag{16}$$

The left large-prime equality residual synthesis is

$$\mathcal{F}^{\mathrm{act,L}}_{h_0} = \sum_{\substack{\gamma \in \mathcal{O}_L,\ j \in \mathbb{Z},\ \rho,\ p \in \mathcal{P}^\sharp_{R_0},\ r \\ pr - m_\rho j = h_0}} \gamma^{(1)}_\rho \mathfrak{A}^{\mathrm{act}}_{\rho,\gamma}(j) \mu(r) \Lambda(pr)\, e_{\gamma, d_\rho r} \otimes e_j. \tag{17}$$

Every coefficient is set to zero off the inherited terminal and orbit support. Thus (17) is finitely supported in j, even though we have embedded it in $\ell^2(\mathbb{Z})$. At h_0 this vector is the negative of the TPC–45/46 synthesis, $-\sum_p U x_p$. It contains literal actual atoms and coefficients, but it belongs to the low-sign-averaged residual Gram route rather than the native affine physical Gram face; the two cross-inner-product structures need not agree.

Define

$$\mathcal{V}_L := \ell^2(\mathcal{O}_L \times \mathcal{S}), \qquad \mathcal{H}_L := \mathcal{V}_L \widehat{\otimes} \ell^2(\mathbb{Z}), \tag{18}$$

where $\mathcal{S}$ contains the supported residual labels. The right space is defined symmetrically, and the full output is their orthogonal sum.

2.2 The frozen-coefficient shift family

For $h \in \mathbb{Z}$, define $\mathcal{F}_h^{\mathrm{fr,L}}$ by the right side of (17) with the determinant equation replaced by

$$pr - m_\rho j = h, \tag{19}$$

while every row, support, residue factor, source coefficient, mask coefficient, and residual label remains the one constructed for h_0. Then

$$\mathcal{F}_{h_0}^{\mathrm{fr,L}} = \mathcal{F}_{h_0}^{\mathrm{act,L}}. \tag{20}$$

The family belongs to $\ell^1(\mathbb{Z}; \mathcal{H}_L) \cap \ell^2(\mathbb{Z}; \mathcal{H}_L)$, and its shift transform is

$$\widehat{\mathcal{F}}^{\mathrm{fr,L}}(\alpha) := \sum_{h \in \mathbb{Z}} \mathcal{F}_h^{\mathrm{fr,L}} \mathrm{e}(-h\alpha). \tag{21}$$

Remark 2.1 (No local-shift provenance). For $h \neq h_0$, $\mathcal{F}_h^{\mathrm{fr,L}}$ is an algebraic extension of one physical coefficient field. Reconstructing TPC at physical shift h would change eligible rows, residue data, the joint multiplier, and possibly the residual label. No uniform identification of those objects is used or claimed.

2.3 Exact projective layers of the full coefficient

TPC–36 proves the pointwise actual-multiplier representation, and TPC–46 inserts the large-prime and Mellin separations. Their combined interface gives the following statement [2, 5].

Proposition 2.2 (Inherited actual-coefficient reassembly)**.** *For each fixed integer $B \geq 0$, the full frozen left family is a Bochner signed superposition of layers*

$$\mathcal{F}_h^{\omega} = \sum_{p,r,m} \sum_{\substack{j \in \mathbb{Z} \\ j \equiv a_\omega \pmod{H_0}}} A_\omega(p,r,m)\phi_\omega(j/J) v_{\omega,p,r,m} \otimes e_j \, \mathbf{1}_{pr-mj=h}, \tag{22}$$

where

(i) *the prime restriction and terminal Mellin factor are retained in A_ω;*

(ii) *$\mu(r)$ is retained in A_ω, while the literal source and mask coefficient is retained in its m-factor;*

(iii) *$v_{\omega,p,r,m} \in \mathcal{V}_L$ is independent of j and is supported on coordinates $(\gamma, d(m)r)$; and*

(iv) *after multiplication by one common fixed compactly supported cutoff that equals one on the physical normalized orbit range, $\phi_\omega \in C_c^\infty(\mathbb{R})$, with*

$$\int \mathfrak{a}_\omega \|\phi_\omega\|_{C^B} \, d|\nu|(\omega) \ll_{B,h_0} X^{o(1)}. \tag{23}$$

Here $\mathfrak{a}_\omega$ is the product of the declared sup-norm weights in the arithmetic and opposite-row factors. More strongly, the construction is coefficientwise before the terminal sum: for every supported m, r,

$$\begin{aligned} &\sum_{\gamma \in \mathcal{O}_L} \gamma_m^{(1)} \mathfrak{A}_{m,\gamma}^{\mathrm{act}}(j) e_{\gamma, d(m)r} \\ &\quad = \sum_{a \bmod H_0} \mathbf{1}_{j \equiv a \pmod{H_0}} \int c_{\omega,a}(m) \phi_{\omega,a}(j/J) q_{\omega,a}^{m,r} \, d\nu_a(\omega), \end{aligned} \tag{24}$$

where

$$q^{m,r}_{\omega,a} = \sum_{\gamma\in\mathcal{O}_L} q_{\omega,a}(\gamma) e_{\gamma,d(m)r}, \tag{25}$$

and the integral of $\|c_{\omega,a}\|_\infty \|q_{\omega,a}\|_\infty \|\phi_{\omega,a}\|_{C^B}$ *is* $X^{o(1)}$. *In particular,*

$$\mathcal{F}^{\mathrm{fr,L}}_h = \int \mathcal{F}^\omega_h \, d\nu(\omega) \tag{26}$$

exactly, with total signed projective mass $X^{o(1)}$ *at every fixed orbit differentiability order.*

Proof. The global row-mask theorem of TPC–36 separates the source row, the opposite row, and a smooth orbit profile after splitting the fixed residue cells. Its fixed-column refinement may additionally split $m = \ell d$, but no such split is needed here. Multiplication by the source coefficient preserves $X^{o(1)}$ projective mass. Mellin inversion separates $\Lambda(pr)$, while the large-prime descent inserts $\mu(r)$ and the literal residual coordinate $d(m)r$. Grouping the opposite-row factor into $v_{\omega,p,r,m}$ gives (22). The inherited orbit factors are smooth on a fixed normalized interval. A common cutoff equal to one there makes them compactly supported without changing a physical coefficient and preserves every fixed seminorm estimate. These are exactly the claims in (23) and (24). Multiplication by the terminal factors and summation over (p, r, m) then gives (26). □

The signed integral in (26) must remain intact. Selecting a favorable ω would not define a bounded projection of the physical multiplier. The orbit projection below is instead a single orthogonal operator applied after the complete sum has been reassembled.

3 The literal fiberwise orbit projection

Let $\mathcal{V}$ be any Hilbert space carrying all non-orbit output labels, and put

$$\widetilde{\mathcal{H}} = \mathcal{V}\widehat{\otimes}\ell^2(\mathbb{Z}). \tag{27}$$

The second tensor factor is the integer orbit coordinate after zero extension. The unitary map

$$\mathscr{F}_j : \ell^2(\mathbb{Z}) \longrightarrow L^2(\mathbb{T}), \qquad (\mathscr{F}_j y)(\beta) = \sum_{j\in\mathbb{Z}} y_j \mathrm{e}(-j\beta), \tag{28}$$

is first defined on finitely supported sequences and then extended by Plancherel.

3.1 The residue-comb band

Fix $H_0 \in \mathbb{N}$ and $0 < \Omega < 1/(2H_0)$. The fixed progression selector has Fourier branches at $H_0^{-1}\mathbb{Z}/\mathbb{Z}$, so the natural low band is the complete residue comb

$$\mathcal{B}^{(H_0)}_\Omega := \left\{\beta \in \mathbb{T} : \mathrm{dist}(\beta, H_0^{-1}\mathbb{Z}) \le \Omega\right\}. \tag{29}$$

It is the disjoint union of H_0 arcs and has measure

$$|\mathcal{B}^{(H_0)}_\Omega| = 2H_0\Omega. \tag{30}$$

Definition 3.1 (Fiberwise orbit-comb projector)**.** On $\widetilde{\mathcal{H}}$, define

$$\Pi^{(H_0)}_\Omega := I_\mathcal{V} \otimes \mathscr{F}_j^{-1} M_{\mathbf{1}_{\mathcal{B}^{(H_0)}_\Omega}} \mathscr{F}_j. \tag{31}$$

The same projection is applied in every opposite-row and residual-label fiber.

Proposition 3.2 (Orthogonality and commutation)**.** *The operator $\Pi_\Omega^{(H_0)}$ is an orthogonal projection of norm one. If S is any bounded operator on $\mathcal{V}$, then*

$$(S \otimes I)\Pi_\Omega^{(H_0)} = \Pi_\Omega^{(H_0)}(S \otimes I). \tag{32}$$

In particular, it commutes with opposite-row coordinate projections, residual-label projections, and the residual Möbius unitary

$$(Uy)_{\gamma,s,j} = \mu(s) y_{\gamma,s,j} \tag{33}$$

on the supported squarefree residual space.

Proof. Multiplication by the indicator of $\mathcal{B}_\Omega^{(H_0)}$ is self-adjoint and idempotent. Unitarity of $\mathscr{F}_j$ proves the first assertion. The two sides of (32) act on separate tensor factors. □

This construction retains the literal product label $s = d(m)r$. Projecting after first merging distinct residual labels would be a different operation and would lose the TPC–45 residual geometry.

3.2 Zero extension and physical compression

Let $I \subset \mathbb{Z}$ be a physical orbit interval of N integers, and let

$$E_I : \mathcal{V}\widehat{\otimes}\ell^2(I) \longrightarrow \widetilde{\mathcal{H}} \tag{34}$$

be zero extension. It is an isometry. The range of $\Pi_\Omega^{(H_0)} E_I$ is the projected sector used below. The compression

$$E_I^* \Pi_\Omega^{(H_0)} E_I \tag{35}$$

is a positive contraction on the finite-orbit-window space, but it is generally not idempotent: the orthogonal projection creates orbit tails outside I. Thus $\Pi_\Omega^{(H_0)}$ is a literal projection in the canonical enlarged space, not a coordinatewise deletion inside the finite window.

For one scalar fiber, the convolution kernel of the projector is

$$\begin{aligned}\kappa_{\Omega,H_0}(n) &:= \int_{\mathcal{B}_\Omega^{(H_0)}} \mathrm{e}(n\beta)\, d\beta \\ &= \left(\sum_{\tau=0}^{H_0-1} \mathrm{e}(n\tau/H_0)\right) \begin{cases} 2\Omega, & n = 0, \\ \dfrac{\sin(2\pi\Omega n)}{\pi n}, & n \neq 0. \end{cases}\end{aligned} \tag{36}$$

In particular, it vanishes unless $H_0 \mid n$, apart from the diagonal convention already contained in the formula.

Theorem 3.3 (Exact physical-window trace)**.** *On one scalar orbit fiber,*

$$\boxed{\mathrm{tr}\big(E_I^* \Pi_\Omega^{(H_0)} E_I\big) = \|\Pi_\Omega^{(H_0)} E_I\|_{\mathrm{HS}}^2 = 2H_0 N\Omega.} \tag{37}$$

For a finite-dimensional non-orbit space of dimension d, the trace is $2dH_0N\Omega$.

Proof. The diagonal matrix coefficient is the measure of the band,

$$\langle \Pi_\Omega^{(H_0)} e_j, e_j \rangle = 2H_0\Omega.$$

Summing over $j \in I$ gives the trace. Orthogonality gives

$$\|\Pi_\Omega^{(H_0)} E_I\|_{\mathrm{HS}}^2 = \mathrm{tr}(E_I^* (\Pi_\Omega^{(H_0)})^2 E_I),$$

which is the same quantity. Tensoring with a finite-dimensional identity repeats the trace in every non-orbit coordinate. □

The number $2H_0N\Omega$ is an exact time–bandwidth trace. The range of $\Pi_\Omega^{(H_0)}$ is infinite dimensional, so this number is not its rank.

3.3 Interaction with the shift transform

For a finitely supported family $(F_h)_h \subset \widetilde{\mathcal{H}}$, boundedness gives

$$\sum_h \Pi_\Omega^{(H_0)} F_h \mathrm{e}(-h\alpha) = \Pi_\Omega^{(H_0)} \sum_h F_h \mathrm{e}(-h\alpha). \tag{38}$$

If

$$F_h = \sum_{p,r,m,j} A_{p,r,m}\phi(j/J) v_{p,r,m} \otimes e_j \, \mathbf{1}_{pr-mj=h}, \tag{39}$$

then, in the orbit Fourier representation,

$$\begin{aligned}&(I_{\mathcal{V}} \otimes \mathscr{F}_j)\widehat{\Pi_\Omega^{(H_0)}} F(\alpha)(\beta)\\ &\quad = \mathbf{1}_{\mathcal{B}_\Omega^{(H_0)}}(\beta) \sum_{p,r,m} A_{p,r,m} v_{p,r,m} \mathrm{e}(-pr\alpha) \sum_j \phi(j/J)\mathrm{e}\big(j(m\alpha-\beta)\big).\end{aligned} \tag{40}$$

The replacement of $m\alpha$ by $m\alpha - \beta$ is the transfer mechanism.

Remark 3.4 (A forbidden commutation). In general

$$\Pi_\Omega^{(H_0)} M_{\phi(j/J)} \neq M_{\phi(j/J)} \Pi_\Omega^{(H_0)}. \tag{41}$$

We apply the projector after the complete coefficient has been multiplied and analyze the resulting convolution. No orbit profile is commuted through the projector.

4 Exact orbit-band to shift-spectrum transfer

We first isolate the support theorem. No primality, Möbius cancellation, or source-mask estimate is used in this section.

Let $0 < M_- \le M_+$, let m range over finitely many integers in $[M_-, M_+]$, and let $\mathcal{V}$ be a Hilbert space. For one residue class $a \pmod{H_0}$, define

$$F_h = \sum_{p,r,m} \sum_{\substack{j\in\mathbb{Z} \\ j\equiv a \pmod{H_0}}} A_{p,r,m}\phi(j/J) v_{p,r,m} \otimes e_j \mathbf{1}_{pr-mj=h}. \tag{42}$$

The arithmetic supports are finite, $v_{p,r,m} \in \mathcal{V}$ is independent of j, and $\phi \in \mathscr{S}(\mathbb{R})$.

4.1 The two-frequency kernel

For $\alpha, \beta \in \mathbb{T}$, set

$$K_{m,a}(\alpha,\beta) := \sum_{\substack{j\in\mathbb{Z} \\ j\equiv a \pmod{H_0}}} \phi(j/J)\mathrm{e}\big(j(m\alpha-\beta)\big). \tag{43}$$

Lemma 4.1 (Progression Poisson formula). *For every $\alpha, \beta \in \mathbb{T}$,*

$$K_{m,a}(\alpha,\beta) = \frac{J}{H_0} \sum_{q\in\mathbb{Z}} \mathrm{e}(aq/H_0)\widehat{\phi}\left(\frac{J}{H_0}\big(q - H_0(m\alpha-\beta)\big)\right). \tag{44}$$

If

$$\operatorname{supp}\widehat{\phi} \subset [-\Sigma, \Sigma], \tag{45}$$

then

$$K_{m,a}(\alpha,\beta) \neq 0 \quad \Longrightarrow \quad \operatorname{dist}\big(m\alpha-\beta, H_0^{-1}\mathbb{Z}\big) \le \frac{\Sigma}{J}. \tag{46}$$

Proof. Write $j = a + H_0 t$ and apply Poisson summation in t. This gives (44). Under (45), a nonzero summand requires

$$|q - H_0(m\alpha - \beta)| \le H_0 \Sigma / J,$$

which is equivalent to (46). $\square$

The shift and orbit Fourier transforms of the projected packet are therefore

$$\begin{aligned}(I_{\mathcal{V}} \otimes \mathscr{F}_j)\widehat{\Pi_\Omega^{(H_0)} F}(\alpha)(\beta) \\ = \mathbf{1}_{\mathcal{B}_\Omega^{(H_0)}}(\beta) \sum_{p,r,m} A_{p,r,m} v_{p,r,m} \mathrm{e}(-pr\alpha) K_{m,a}(\alpha, \beta).\end{aligned} \tag{47}$$

4.2 The exact projected gap

Put

$$W := \Omega + \frac{\Sigma}{J}. \tag{48}$$

Theorem 4.2 (Exact orbit-band transfer)**.** *Assume* (45) *and*

$$0 < W < \frac{1}{2H_0}. \tag{49}$$

Then

$$\boxed{\widehat{\Pi_\Omega^{(H_0)} F}(\alpha) = 0} \tag{50}$$

whenever

$$\boxed{\frac{W}{M_-} < \|\alpha\|_{\mathbb{T}} < \frac{H_0^{-1} - W}{M_+}.} \tag{51}$$

The assertion is uniform in the coefficients and Hilbert vectors in (42)*. The assertion is vacuous unless*

$$\frac{W}{M_-} < \frac{H_0^{-1} - W}{M_+}. \tag{52}$$

Proof. Let $t = \|\alpha\|_{\mathbb{T}}$ and choose a representative with $|\alpha| = t$. For every supported m,

$$W < mt < H_0^{-1} - W. \tag{53}$$

If $\beta \in \mathcal{B}_\Omega^{(H_0)}$, choose q/H_0 with $|\beta - q/H_0| \le \Omega$. The finite set $H_0^{-1}\mathbb{Z}/\mathbb{Z}$ is a subgroup of $\mathbb{T}$, so

$$\begin{aligned}\operatorname{dist}(m\alpha - \beta, H_0^{-1}\mathbb{Z}) &\ge \operatorname{dist}(m\alpha, H_0^{-1}\mathbb{Z}) - \Omega \\ &> W - \Omega = \Sigma/J.\end{aligned} \tag{54}$$

Here (53) places $m\alpha$ strictly between the two nearest subgroup points 0 and $\operatorname{sgn}(\alpha)/H_0$. Thus (46) fails for every m and almost every β retained by the projector. Formula (47) is zero in $L^2(\mathcal{B}_\Omega^{(H_0)}; \mathcal{V})$, proving the theorem. $\square$

Corollary 4.3 (Signed projective superpositions)**.** *Let $F = \int F^\omega \, d\nu(\omega)$, where every layer has the form* (42)*, the same $M_\pm, H_0, J$, and Fourier bandwidth at most Σ. If the integral is absolutely convergent in $\ell^1(\mathbb{Z}; \widetilde{\mathcal{H}})$, then* (50) *holds for the full signed superposition on* (51)*.*

Proof. The projected transform of each layer is zero on the same interval. Boundedness of the projector and absolute convergence permit the Bochner integral to pass through both Fourier transforms. $\square$

Corollary 4.4 (Vector-valued coefficient band)**.** *For each* (p, r, m)*, let* $g_{p,r,m} \in \ell^2(\mathbb{Z}; \mathcal{V})$ *satisfy*

$$\operatorname{supp} \mathscr{F}_j g_{p,r,m} \subset \mathcal{B}^{(H_0)}_{\Sigma/J}. \tag{55}$$

Define

$$F_h = \sum_{p,r,m,j} A_{p,r,m} g_{p,r,m}(j) \otimes e_j \mathbf{1}_{pr-mj=h}. \tag{56}$$

Then (50) *holds on* (51)*, with* $W = \Omega + \Sigma/J$*. The shift transform is understood in* $L^2(\mathbb{T}; \widetilde{\mathcal{H}})$*; because the arithmetic support is finite, the modal expression also converges in* $\widetilde{\mathcal{H}}$ *for each* α*.*

Proof. In the orbit Fourier representation, the (p, r, m)-term at output frequency β is $\widehat{g}_{p,r,m}(\beta - m\alpha)$. It can meet $\mathcal{B}^{(H_0)}_{\Omega}$ only if

$$d_{H_0}(m\alpha) \le \Omega + \Sigma/J.$$

The interval (51) excludes this for every supported m, exactly as in the proof of theorem 4.2. □

Remark 4.5 (Amplitude, not atomic energy). The vanishing concerns the Hilbert-valued amplitude $h \mapsto \Pi^{(H_0)}_{\Omega} F_h$. It does not imply a gap for $h \mapsto \|F_h\|^2$, because squaring convolves shift frequencies. Nor does it compare the synthesis at one shift with the atomic diagonal at that shift.

Remark 4.6 (Why the whole residue comb is kept). The selector $\mathbf{1}_{j \equiv a \pmod{H_0}}$ has all H_0 Fourier branches. Projecting only near zero would define a smaller valid sector, but it would suppress the other inherited residue modes. The comb (29) retains them all and changes the time–bandwidth trace only by the fixed factor H_0.

5 Smooth profiles: rapid leakage and a fixed core

The actual orbit profiles are smooth, not exactly band limited. We give two complementary formulations: a direct projected Poisson bound and an exact low-band core with a quantified remainder.

5.1 Compact-support provenance

The required endpoint regularity is stronger than smooth values on a finite set. A hard cutoff in j would create Dirichlet tails.

Lemma 5.1 (Inherited common orbit support)**.** *The orbit profiles in proposition 2.2 may be chosen in one fixed compact subset* $K \subset \mathbb{R}$*, independently of* X *and of the projective parameter. For every fixed* B*, the signed projective mass of their* $C^B(K)$ *norms is* $X^{o(1)}$*.*

Proof. The provenance theorem of TPC–25 places every row-dependent smooth orbit product, in the normalized variable $t = j/J$, in a fixed bounded subset of $C^\infty_c(\mathbb{R})$, and its Mellin reassembly has integrable polynomial seminorms [1]. TPC–36 imports that residue/Mellin representation and proves that multiplication by the actual static mask does not change the orbit seminorm mass [2]. The large-prime and residual rewrites in TPC–45 and TPC–46 change arithmetic and Hilbert labels but do not modify the orbit profile. This proves the statement. □

This lemma is the point at which the physical smooth localization is used. Without it, arbitrary-power leakage below would be false.

5.2 Direct smooth decay

For $\theta \in \mathbb{T}$, write

$$d_{H_0}(\theta) := \operatorname{dist}(\theta, H_0^{-1}\mathbb{Z}). \tag{57}$$

Lemma 5.2 (Projected progression profile). *Let $\phi \in C_c^\infty(K)$. For every integer $B \geq 1$,*

$$\begin{aligned} &\left\| \Pi_\Omega^{(H_0)} \left(\mathbf{1}_{j \equiv a \pmod{H_0}} \phi(j/J) \mathrm{e}(mj\alpha) \right) \right\|_{\ell^2(\mathbb{Z})} \\ &\qquad \ll_{B,H_0,K} (\Omega J^2)^{1/2} \Big(1 + J\big(d_{H_0}(m\alpha) - \Omega\big)_+\Big)^{-B} \|\phi\|_{C^{B+2}}. \end{aligned} \tag{58}$$

Equivalently, if $\Omega = \Lambda/J$, the leading factor is $O_{H_0}((\Lambda J)^{1/2})$.

Proof. In the orbit Fourier representation the squared norm is the integral, over $\beta \in \mathcal{B}_\Omega^{(H_0)}$, of the squared modulus of $K_{m,a}(\alpha, \beta)$ from (43). Repeated integration by parts gives

$$|\widehat{\phi}(\xi)| \ll_{B,K} (1 + |\xi|)^{-B-2} \|\phi\|_{C^{B+2}}.$$

Insert this in (44) and sum the integer translates. Uniformly in α, β,

$$|K_{m,a}(\alpha, \beta)| \ll_{B,H_0,K} J \left(1 + J d_{H_0}(m\alpha - \beta)\right)^{-B} \|\phi\|_{C^{B+2}}. \tag{59}$$

For β in the comb band,

$$d_{H_0}(m\alpha - \beta) \geq \big(d_{H_0}(m\alpha) - \Omega\big)_+.$$

Integrating (59) over a set of measure $2H_0\Omega$ proves (58). □

For the packet (42), define the coefficient envelope

$$\mathcal{M}_1(F) := \sum_{p,r,m} |A_{p,r,m}| \|v_{p,r,m}\|_{\mathcal{V}}. \tag{60}$$

Corollary 5.3 (Direct smooth packet bound). *Under the hypotheses of lemma 5.2,*

$$\begin{aligned} \|\widehat{\Pi_\Omega^{(H_0)} F}(\alpha)\| &\ll_{B,H_0,K} (\Omega J^2)^{1/2} \mathcal{M}_1(F) \|\phi\|_{C^{B+2}} \\ &\quad \times \max_{M_- \leq m \leq M_+} \left(1 + J(d_{H_0}(m\alpha) - \Omega)_+\right)^{-B}. \end{aligned} \tag{61}$$

Proof. Use the modal factorization (47), the triangle inequality in $L^2(\mathcal{B}_\Omega^{(H_0)}; \mathcal{V})$, and lemma 5.2. □

5.3 A fixed band-limited core

Choose once and for all $\chi \in C_c^\infty(\mathbb{R})$ satisfying

$$0 \leq \chi \leq 1, \qquad \chi(\xi) = 1 \quad (|\xi| \leq 1/2), \qquad \chi(\xi) = 0 \quad (|\xi| \geq 1). \tag{62}$$

For $J \geq 1$ and $\Sigma \geq 2$, define

$$\widehat{\phi_{\leq \Sigma}}(\xi) := \chi(\xi/\Sigma)\widehat{\phi}(\xi), \qquad\qquad \phi_{>\Sigma} := \phi - \phi_{\leq \Sigma}. \tag{63}$$

The low component has outer Fourier half-width exactly at most Σ, so theorem 4.2 applies with $W = \Omega + \Sigma/J$.

Lemma 5.4 (Sampled high-frequency tail). *Assume $J \geq 1$. For every integer $B \geq 2$, uniformly over $\phi \in C_c^\infty(K)$,*

$$\left(\sum_{j\in\mathbb{Z}} |\phi_{>\Sigma}(j/J)|^2\right)^{1/2} \ll_{B,K} J^{1/2}\Sigma^{1/2-B}\|\phi\|_{C^{B+2}}. \tag{64}$$

The same estimate holds after restriction to any progression modulo H_0.

Proof. Repeated integration by parts gives $|\widehat{\phi}(\xi)| \ll_{B,K} (1+|\xi|)^{-B}\|\phi\|_{C^B}$. Plancherel and the support of $1-\chi(\xi/\Sigma)$ therefore give

$$\|\phi_{>\Sigma}\|_{L^2} + \|\phi'_{>\Sigma}\|_{L^2} \ll_{B,K} \Sigma^{1/2-B}\|\phi\|_{C^{B+2}}. \tag{65}$$

For every $g \in H^1(\mathbb{R})$, the elementary sampling inequality on the intervals $[j/J,(j+1)/J]$ is

$$\sum_{j\in\mathbb{Z}} |g(j/J)|^2 \ll J\big(\|g\|_{L^2}^2 + \|g'\|_{L^2}^2\big). \tag{66}$$

Apply this to $g = \phi_{>\Sigma}$ and use (65). A progression is a subsequence, so its sum is no larger. □

Theorem 5.5 (Exact core plus controlled leakage). *Let $F = F_{\leq\Sigma} + F_{>\Sigma}$ be obtained from (42) by applying (63). On the exact gap*

$$\frac{\Omega+\Sigma/J}{M_-} < \|\alpha\|_{\mathbb{T}} < \frac{H_0^{-1}-\Omega-\Sigma/J}{M_+}, \tag{67}$$

one has

$$\widehat{\Pi_\Omega^{(H_0)} F_{\leq\Sigma}}(\alpha) = 0, \tag{68}$$

$$\|\widehat{\Pi_\Omega^{(H_0)} F}(\alpha)\| \leq C_{B,H_0,K} J^{1/2}\Sigma^{1/2-B}\mathcal{M}_1(F)\|\phi\|_{C^{B+2}}. \tag{69}$$

Proof. The first identity is theorem 4.2. Hence the full projected transform equals the projected transform of the high component. Modulation by $\mathrm{e}(mj\alpha)$ is unitary on $\ell^2(\mathbb{Z})$, and the projector is contractive. Apply the triangle inequality over (p,r,m), followed by lemma 5.4. □

The tail estimate is deliberately expressed relative to the declared envelope $\mathcal{M}_1(F)$. It contains no unproved cancellation among the arithmetic coefficients.

6 Transfer to the full actual-coefficient residual packet

We apply the output projector $\Pi_\Omega^{(H_0)}$ to the fully reassembled coefficient field. The auxiliary coefficient projector $Q_\Sigma^{(H_0)}$ below supplies only a proof decomposition, while the projective representation is used only to prove regularity; neither defines the projected output sector.

6.1 A representation-independent coefficient cutoff

For a supported source row $m = \ell d(m)$ and cofactor r, define the left vector profile

$$G_{m,r}^L(j) := \sum_{\gamma\in\mathcal{O}_L} \gamma_m^{(1)}\mathfrak{A}_{m,\gamma}^{\mathrm{act}}(j)e_{\gamma,d(m)r} \in \mathcal{V}_L. \tag{70}$$

It is zero outside the physical orbit support. Then the frozen family from section 2 is

$$\mathcal{F}_h^{\mathrm{fr,L}} = \sum_{p,r,m,j} \mu(r)\Lambda(pr)G_{m,r}^L(j)\otimes e_j\mathbf{1}_{pr-mj=h}. \tag{71}$$

Let

$$Q_\Sigma^{(H_0)} := \mathscr{F}_j^{-1} M_{\mathbf{1}_{\mathcal{B}^{(H_0)}_{\Sigma/J}}} \mathscr{F}_j \tag{72}$$

act on the j-variable of every $\mathcal{V}_L$-valued profile. Put

$$G^{\mathrm{lo}}_{m,r} := Q_\Sigma^{(H_0)} G^L_{m,r}, \qquad G^{\mathrm{hi}}_{m,r} := (I - Q_\Sigma^{(H_0)}) G^L_{m,r}. \tag{73}$$

This is a canonical orthogonal split of the full actual coefficient. It is independent of any choice of projective representation. Because $Q_\Sigma^{(H_0)}$ is a sharp Fourier projection, $G^{\mathrm{lo}}_{m,r}$ can have sinc tails and need not belong to $\ell^1(\mathbb{Z};\mathcal{V}_L)$. All low/high frozen families below are therefore defined in ℓ^2, and their shift transforms use the unitary partial Fourier transform. The arithmetic support in (p,r,m) is finite, so the modal sums converge in the output Hilbert space for each α; no absolute j-summability of the sharp core is asserted. Throughout this subsection we assume

$$0 < \Sigma/J < 1/(2H_0), \tag{74}$$

so the arcs defining the coefficient comb are disjoint.

Proposition 6.1 (Full-profile Fourier tail)**.** *For every fixed integer $B \geq 2$,*

$$\sup_{m,r} \|G^{\mathrm{hi}}_{m,r}\|_{\ell^2(\mathbb{Z};\mathcal{V}_L)} \ll_{B,h_0} Q^{1/2} J^{1/2} X^{o(1)} \Sigma^{1/2-B}. \tag{75}$$

The symmetric estimate holds for the right profile.

Proof. Apply the coefficientwise identity (24), including its full signed integral, to (70). For each layer, lemma 5.4 constructs a progression-band approximant with error

$$\ll_{B,h_0} J^{1/2} \Sigma^{1/2-B} \|\phi_\omega\|_{C^{B+2}}.$$

The orthogonal projection $Q_\Sigma^{(H_0)}$ is the best ℓ^2 approximation from its range, so its error is no larger. Minkowski's inequality reassembles all signed layers. The opposite-row vector has at most $QX^{o(1)}$ supported coordinates; its ℓ^2 norm is therefore bounded by $Q^{1/2}X^{o(1)}$ times the inherited sup norm. Finally use the projective seminorm mass (23). The residual coordinate $d(m)r$ is a single basis label and contributes no additional norm. □

This proof neither chooses a favorable projective layer nor replaces the source mask by its absolute value. The absolute projective mass is used only after the exact signed identity has been inserted.

6.2 The actual-coefficient projected theorem

Let $\mathcal{F}^{\mathrm{fr,L}}_{h,\mathrm{lo}}$ and $\mathcal{F}^{\mathrm{fr,L}}_{h,\mathrm{hi}}$ be obtained from (71) by using the profiles in (73).

Theorem 6.2 (Projected actual-coefficient spectral transfer)**.** *For every fixed integer $B \geq 2$, assume $0 < \Omega + \Sigma/J < 1/(2H_0)$. On*

$$\frac{\Omega + \Sigma/J}{M_-} < \|\alpha\|_{\mathbb{T}} < \frac{H_0^{-1} - \Omega - \Sigma/J}{M_+}, \tag{76}$$

one has

$$\Pi_\Omega^{(H_0)} \widehat{\mathcal{F}^{\mathrm{fr,L}}_{\mathrm{lo}}}(\alpha) = 0, \tag{77}$$

$$\left\| \Pi_\Omega^{(H_0)} \widehat{\mathcal{F}^{\mathrm{fr,L}}}(\alpha) \right\| \ll_{B,h_0} X^{867/400+o(1)} \Sigma^{1/2-B}. \tag{78}$$

For one fixed opposite output row, the exponent $867/400$ *in* (78) *may be replaced by* $1467/800$*. The same conclusions hold on the right and hence on the orthogonal sum of both sides.*

Proof. The low profile satisfies

$$\operatorname{supp} \mathscr{F}_j G^{\mathrm{lo}}_{m,r} \subset \mathcal{B}^{(H_0)}_{\Sigma/J}.$$

Thus (77) is corollary 4.4. On the same gap, the full projected transform equals the high-profile contribution. For each (p, r, m), modulation in j and $\Pi^{(H_0)}_{\Omega}$ have norm one. The triangle inequality and proposition 6.1 give

$$\left\| \widehat{\Pi^{(H_0)}_{\Omega} \mathcal{F}^{\mathrm{fr,L}}}(\alpha) \right\| \ll X^{o(1)} \#\{(p, r, m)\} Q^{1/2} J^{1/2} \Sigma^{1/2-B}. \tag{79}$$

The terminal logarithm is included in $X^{o(1)}$, and the supported product pairs and source rows satisfy

$$\#\{(p, r)\} \le X^{1+o(1)}, \qquad \#\{m\} \le Q X^{o(1)}. \tag{80}$$

Since $Q = X^{267/400+o(1)}$, $J = X^{133/400+o(1)}$, the total exponent is

$$1 + \frac{3}{2}\frac{267}{400} + \frac{1}{2}\frac{133}{400} = \frac{867}{400}. \tag{81}$$

Fixing one opposite row removes the factor $Q^{1/2}$, leaving

$$1 + \frac{267}{400} + \frac{1}{2}\frac{133}{400} = \frac{1467}{800}. \tag{82}$$

This proves all assertions. ☐

Corollary 6.3 (Arbitrary-power actual leakage)**.** *In the endpoint regime $J = X^{\nu+o(1)}$, fix $0 < \kappa < \nu$ and take $\Sigma = X^{\kappa+o(1)}$. For every $A > 0$, one may choose one fixed differentiability order B, depending only on A, κ, h_0, such that*

$$\left\| \widehat{\Pi^{(H_0)}_{\Omega} \mathcal{F}^{\mathrm{fr,L/R}}}(\alpha) \right\| \ll_{A,\kappa,h_0} X^{-A} \tag{83}$$

uniformly on (76).

Proof. In (78), choose $B > 1/2 + (A + 867/400 + 1)/\kappa$. The fixed-order projective seminorm estimate remains $X^{o(1)}$, which is absorbed for large X. ☐

6.3 Residual synthesis versus the grouped physical column

At $h = h_0$, the residual vector in (71) is precisely the negative large-prime synthesis $-\sum_p U x_p$ of TPC–45/46. It uses the literal actual coefficients and signs. It belongs, however, to the low-sign-averaged residual Gram route; it is not the native affine physical face, whose cross inner products are different [4, 5].

For each opposite row γ, let

$$\mathcal{S}_\gamma := \{s : (\gamma, s) \text{ occurs at some physical orbit time}\} \tag{84}$$

and restrict to the orbit-invariant space

$$\mathcal{H}_{\mathrm{act,res}} := \bigoplus_{\gamma \in \mathcal{O}_L} \ell^2(\mathcal{S}_\gamma) \widehat{\otimes} \ell^2(\mathbb{Z}). \tag{85}$$

For comparison, define the residual grouping map on this space by

$$(Cy)_{\gamma,j} := \sum_s y_{\gamma,s,j}. \tag{86}$$

Proposition 6.4 (Grouped large-prime component)**.** *The map C commutes with the orbit projector. If*

$$\rho_{\mathrm{res}} := \max_{\gamma} |\mathcal{S}_\gamma|, \tag{87}$$

then

$$\|C\|^2 = \rho_{\mathrm{res}}. \tag{88}$$

At h_0, grouping $-\sum_p Ux_p$ recovers exactly the corresponding large-prime component of the actual equality column under the sign convention of section 2. Exact core vanishing survives grouping, while the leakage bound acquires at most $\rho_{\mathrm{res}}^{1/2}$.

Proof. The domain (85) is invariant under the orbit projector. The two operators act in the residual and orbit tensor factors, respectively. Fiberwise Cauchy–Schwarz gives $\|C\|^2 \le \rho_{\mathrm{res}}$, and a constant vector on a largest fiber attains equality. The physical identification at h_0 is the definition of the residual large-prime lift. Apply C to (77) and (78). □

This proposition does not restore the base component or the full native affine Gram form. The grouping norm may itself be polynomial, so it is not a fixed-shift closure theorem.

7 The endpoint exponent ledger

Write the source and orbit exponents as

$$M_- \asymp M_+ \asymp X^{\mu}, \qquad J \asymp X^{\nu}, \qquad \mu + \nu = 1. \tag{89}$$

At the TPC endpoint,

$$\mu = \frac{267}{400}, \qquad \nu = \frac{133}{400}. \tag{90}$$

7.1 A one-parameter family of projected sectors

Fix

$$0 < \kappa < \nu, \qquad \Omega = \frac{X^{\kappa+o(1)}}{J}, \qquad \Sigma = X^{\kappa+o(1)}. \tag{91}$$

Then

$$W = \Omega + \Sigma/J = X^{\kappa-\nu+o(1)}. \tag{92}$$

The lower endpoint in (76) is

$$\frac{W}{M_-} = X^{-1+\kappa+o(1)}, \tag{93}$$

while its upper endpoint is

$$\frac{H_0^{-1} - W}{M_+} = X^{-\mu+o(1)}. \tag{94}$$

Theorem 7.1 (Robust projected endpoint range)**.** *Under (89) and (91), fix*

$$0 < \epsilon < \frac{\nu - \kappa}{2}. \tag{95}$$

For all sufficiently large X, the interval

$$\boxed{X^{-1+\kappa+\epsilon} \le \|\alpha\|_{\mathbb{T}} \le X^{-\mu-\epsilon}} \tag{96}$$

is contained in (76). The ideal coefficient core vanishes there exactly, and the projected frozen family built from the literal physical h_0-coefficient field obeys (83).

Proof. The lower endpoint in (96) exceeds (93) by $X^{\epsilon+o(1)}$. At the upper endpoint, $M_+\|\alpha\|_{\mathbb{T}} \le X^{-\epsilon+o(1)}$, so it is smaller than $H_0^{-1} - W$. The interval is nonempty precisely when

$$-1 + \kappa + \epsilon < -\mu - \epsilon,$$

which is (95). Apply theorem 6.2 and corollary 6.3. □

7.2 The TPC allowance $\kappa = 1/400$

Take

$$\kappa = \frac{1}{400}. \tag{97}$$

Then

$$\nu - \kappa = \frac{132}{400} = \frac{33}{100}. \tag{98}$$

Corollary 7.2 (The TPC–47 mesoscopic sector)**.** *For every*

$$0 < \epsilon < \frac{33}{200}, \tag{99}$$

the projected frozen shift family built from the literal physical h_0-coefficient field has arbitrary-power shift-spectral decay on

$$\boxed{X^{-399/400+\epsilon} \leq \|\alpha\|_{\mathbb{T}} \leq X^{-267/400-\epsilon}.} \tag{100}$$

Its canonical low coefficient core vanishes exactly on the larger open interval (76)*.*

The fine-frequency endpoint in TPC–46 was $X^{-1+o(1)}$. The actual-residual-output projection costs $1/400$ in that endpoint but leaves the strict exponent width $33/100$. This is a transfer loss, not an estimate for the fixed physical atom.

7.3 Time–bandwidth and cover count

Let $N \asymp J$ be the length of the physical orbit interval. From theorem 3.3,

$$\operatorname{tr}(E_I^* \Pi_\Omega^{(H_0)} E_I) = 2H_0 N \Omega = X^{\kappa+o(1)}. \tag{101}$$

The comb occupies the frequency fraction

$$|\mathcal{B}_\Omega^{(H_0)}| = 2H_0\Omega = X^{\kappa-\nu+o(1)}. \tag{102}$$

Thus covering the circle by comparable translates requires

$$X^{\nu-\kappa+o(1)} \tag{103}$$

bands. At $\kappa = 1/400$, the three exact scales are

quantity	scale
time–bandwidth trace per residual fiber	$X^{1/400+o(1)}$
projected shift-gap exponent width	$33/100$
number of comparable orbit bands covering $\mathbb{T}$	$X^{33/100+o(1)}$.

(104)

The cover count does not prove that summing all bands incurs that loss; coefficients from different bands may interact. It records only the geometry that a single projected theorem leaves unresolved.

7.4 Rational identities used later

For reproducibility, the endpoint ledger is

$$\frac{267}{400} + \frac{133}{400} = 1, \tag{105}$$

$$1 - \frac{1}{400} = \frac{399}{400}, \tag{106}$$

$$\frac{133}{400} - \frac{1}{400} = \frac{33}{100}, \tag{107}$$

$$\frac{1}{2}\left(\frac{33}{100}\right) = \frac{33}{200}, \tag{108}$$

$$1 + \frac{3}{2}\frac{267}{400} + \frac{1}{2}\frac{133}{400} = \frac{867}{400}, \tag{109}$$

$$1 + \frac{267}{400} + \frac{1}{2}\frac{133}{400} = \frac{1467}{800}. \tag{110}$$

8 The retained-mass frontier

The projected sector is not the zero subspace on a physical orbit window. Nevertheless, the projector can retain an arbitrarily small fraction of an ambient finite-orbit vector. No such smallness is asserted for the actual arithmetic packet. Both operator-theoretic statements are exact.

8.1 Qualitative injectivity on every finite window

Theorem 8.1 (Finite-support injectivity)**.** *Let $0 < \Omega < 1/(2H_0)$. If $y \in \mathcal{V}\widehat{\otimes}\ell^2(\mathbb{Z})$ is finitely supported in the orbit coordinate, then*

$$\Pi_\Omega^{(H_0)} y = 0 \quad \Longrightarrow \quad y = 0. \tag{111}$$

Equivalently, for every finite interval I, the compression

$$E_I^* \Pi_\Omega^{(H_0)} E_I \tag{112}$$

is strictly positive definite.

Proof. Pair y with an arbitrary vector in $\mathcal{V}$. Its orbit Fourier transform is a Laurent polynomial in $z = \mathrm{e}(-\beta)$. If the projection is zero, this polynomial vanishes almost everywhere on every open arc in $\mathcal{B}_\Omega^{(H_0)}$, hence identically. Every scalar pairing is zero, so all coefficients of y vanish. Applying this to $E_I y$ proves strict positivity of the compression. ☐

Thus a nonzero finitely supported residual packet always has a nonzero projected component. This is only a qualitative statement; the smallest eigenvalue of (112) is not bounded below uniformly in the window length.

8.2 An exponentially thin finite-support family

Let S be the unit shift on $\ell^2(\mathbb{Z})$, $(Sy)_j = y_{j-H_0}$, and for $n \geq 1$ put

$$f_n := (I - S)^n e_0 = \sum_{k=0}^{n} (-1)^k \binom{n}{k} e_{kH_0}. \tag{113}$$

Then

$$(\mathscr{F}_j f_n)(\beta) = \big(1 - \mathrm{e}(-H_0\beta)\big)^n, \tag{114}$$

$$\|f_n\|_2^2 = \binom{2n}{n}. \tag{115}$$

Theorem 8.2 (No uniform retained-mass bound)**.** *There is an absolute constant C such that, whenever $\pi H_0 \Omega < 1$,*

$$0 < \frac{\|\Pi_\Omega^{(H_0)} f_n\|_2}{\|f_n\|_2} \le C n^{1/4} (H_0 \Omega)^{1/2} (\pi H_0 \Omega)^n. \tag{116}$$

The support of f_n lies in an interval of length $nH_0 + 1$, so it may be translated into any physical orbit window of at least that length.

Proof. Write $\beta = \tau/H_0 + u$ on one component of the comb, with $|u| \le \Omega$. Then

$$|1 - \mathrm{e}(-H_0\beta)| = 2|\sin(\pi H_0 u)| \le 2\pi H_0 \Omega. \tag{117}$$

There are H_0 components, with total measure $2H_0\Omega$. Hence

$$\|\Pi_\Omega^{(H_0)} f_n\|_2^2 \le 2H_0\Omega(2\pi H_0\Omega)^{2n}. \tag{118}$$

The elementary central-binomial estimate

$$\binom{2n}{n} \ge \frac{4^n}{2\sqrt{n}} \tag{119}$$

and (115) now give (116), after changing an absolute constant. Strict positivity follows from theorem 8.1. □

At the endpoint $\Omega = X^{\kappa-\nu+o(1)}$, for any proposed fixed power C_0, choosing one fixed n large enough makes the ratio in (116) smaller than X^{-C_0} for all large X. Thus there is no uniform polynomial inverse

$$\|y\|_2 \le X^{C_0} \|\Pi_\Omega^{(H_0)} y\|_2 \tag{120}$$

on orbit windows of length J.

8.3 Consequences for the TPC gate

The two theorems above establish the exact logical status of the sector:

(i) it is nonzero on every nonzero finite-orbit vector, hence is not vacuous if the finite physical residual packet is nonzero;

(ii) on ambient finite-orbit vectors it need not retain a fixed, logarithmic, or even polynomially bounded fraction of the norm; no corresponding smallness is asserted for the actual arithmetic packet; and

(iii) the trace identity (37) is an average spectral-size statement, not a lower bound for one arithmetic vector.

There is also no coordinatewise domination. For example, $\Pi_\Omega^{(H_0)} e_0$ is the sinc-comb sequence (36); it has nonzero coordinates away from zero although e_0 does not. Only the global Hilbert contraction

$$\|\Pi_\Omega^{(H_0)} y\|_2 \le \|y\|_2 \tag{121}$$

is automatic.

Remark 8.3 (Why arbitrary dual fields restore the obstruction)*.* Before projection, modulation by $\mathrm{e}(mj\alpha)$ is unitary in the orbit coordinate. An arbitrary dual vector may be oppositely modulated to place its mass at any prescribed shift frequency. The full output unit ball therefore has no uniform gap derived solely from smoothness of the source mask. The theorem succeeds precisely because it states and retains the orbit-band restriction.

9 Boundaries and the next gate

The positive result is a transfer theorem for one canonical family of orbit bands. This section records every identification that would be needed before it could enter a fixed-shift prime-pair argument.

9.1 Five distinctions that remain active

Residual synthesis versus the native affine face. At h_0, the vector analyzed here is the actual-coefficient negative large-prime residual synthesis $-\sum_p U x_p$. TPC–45 obtains this object after averaging the lower prime signs. Its primewise norms agree with the corresponding native vectors, but its cross inner products do not generally agree with those of the physical affine face [4]. The grouping statement in proposition 6.4 recovers only the large-prime component and pays the exact residual-fiber norm.

Frozen shifts versus reconstructed physical shifts. Only $h = h_0$ has inherited physical provenance. The shift transform uses every integer h after freezing the h_0 coefficient field. For a newly reconstructed physical shift, the row set, residue factor, mask, and source residues may change. No growing-window uniformity theorem is asserted.

Amplitude spectrum versus atomic normalization. Theorems 4.2 and 6.2 concern the Fourier transform of a Hilbert amplitude. Fourier inversion at one h_0 integrates all frequencies, including the uncontrolled complement. Moreover, the TPC denominator is the atomic diagonal at h_0, not the Cartesian energy over all determinant shifts. TPC–46 proves that these normalizations may differ by a polynomial factor [5].

Consecutive completion versus sparse aliases. The actual TPC completion samples determinant intercepts separated by the triple-prime modulus. Sparse sampling folds the completed shift spectrum. The present orbit projection neither unfolds those aliases nor proves that the equality-alias residual $d(m)r$ remains literal at nonzero aliases; there the squarefree kernel can replace the product.

One band family versus the full orbit space. The complement $I - \Pi_\Omega^{(H_0)}$ is uncontrolled. By theorem 8.2, qualitative injectivity of the low sector supplies no quantitative reconstruction. Summing separate pointwise bounds over the $X^{33/100+o(1)}$ comparable bands in (103) would introduce a prohibitive support factor unless orthogonality is retained through the later operations.

9.2 What has genuinely changed

Before this paper, the orbit-output coupling was an arbitrary j-dependent Hilbert vector, so the TPC–46 spectral theorem could not be applied to any declared portion of the actual residual field. Now there is a representation-independent norm-one operator, applied to every output row and residual label, for which

(i) the full actual coefficient reassembles with its source mask and Möbius roles intact;

(ii) a growing $X^{1/400+o(1)}$ time–bandwidth sector has super-polynomial leakage on the interval (100); and

(iii) the exact operator-theoretic obstruction to uniform recovery is proved, rather than hidden in a phrase such as "smooth output."

This crosses Door A of TPC–46 on one bounded projected residual sector. It does not cross the other three doors.

9.3 A reversible continuation

The next paper may pursue any of the following logically separate tasks.

Door A. Partition the entire orbit circle into translated comb bands and prove a two-parameter resonance-tube square-function estimate that preserves orthogonality while the bands are reassembled. This avoids asking one low band to contain a fixed fraction of the vector.

Door B. Use the literal TPC source/mask phases to prove a coefficient-specific spectral concentration or entropy bound. It must beat the arbitrary-vector counterexample in theorem 8.2 by using arithmetic data.

Door C. Combine orbit-band projection with the exact residual grouping map and prove that one additional cofactor range has acceptable grouping multiplicity at atomic normalization.

Door D. Construct a family of actual masks or residual packets that saturates the high-frequency or grouping frontier. This would give a sharp negative boundary without asserting failure of every possible TPC continuation.

Door A is the most direct continuation: the projectors are orthogonal before reassembly, and the bandwise resonance condition is explicit. The missing theorem is a square-function or large-sieve estimate that survives determinant dilation, residual grouping, and fixed-shift localization. No such estimate is assumed here.

9.4 Prohibited conclusions

Nothing in this paper proves or implies

(i) deterministic cancellation of Möbius coefficients at the prescribed physical shift;

(ii) an atomic-normalized Bessel bound for the complete equality column;

(iii) a uniform family of actual TPC packets over growing shifts;

(iv) a parity-barrier breach, the Hardy–Littlewood prime-pair asymptotic, or the twin-prime conjecture.

The advance is sectorial: the canonical low core has an exact gap, while the projected frozen family built from the literal h_0-coefficient field has arbitrary-power leakage there. Its possible quantitative thinness is also proved.

10 Exact certificate and claim ledger

The reproducibility directory contains

`experiments/tpc47_certificate.py`.

It uses only the Python standard library and writes a canonical JSON report. A successful run performs 28,428 checks. Its canonical payload digest is

`9e01380fae0d684d96b6ed23e5f99b4b`
`a8359667d6182459e36eeaeb1f66fdba`.

The normalized source SHA–256 is

```
719fd93d61b75ade58d4817b0b870134
3060cadb87c9c9f4aad38c521ad8e356,
```

and the JSON SHA–256 is

```
6b0d0c41b335e437affd5d06f28495f
09655e265144253f196d4af9396b17e19.
```

These values are also recorded in the repository README.

10.1 What the finite certificate checks

The main exact model is on $\mathbb{Z}/60\mathbb{Z}$ over $\mathbb{F}_{601}$, which contains a primitive sixtieth root of unity. It retains

$$H_0 = 3, \qquad \text{the literal static mask}, \qquad s = d(m)r, \tag{122}$$

as well as separate opposite-row and residual fibers. The certificate verifies:

(i) exact finite Fourier inversion and fiberwise projector idempotence;

(ii) the three residue-comb branches created by $j \equiv a \pmod 3$;

(iii) the support convolution for a projected output field and a low orbit profile;

(iv) the determinant dilation identity $K_m(k) = \widehat{g}(-mk)$ and the resulting exact finite gap;

(v) direct synthesis versus modal factorization for a packet that retains the actual-mask matrix and the label $s = d(m)r$;

(vi) whole signed projective reassembly and the exact low-plus-tail identity, without selecting one layer;

(vii) finite-difference Fourier identities and Hilbert contraction;

(viii) exact finite Fourier rank boundaries, rational endpoint identities, and explicit failure witnesses.

The cyclic calculation is a finite analogue of theorem 4.2; it is not a numerical proof of the continuous Poisson theorem. The latter is proved directly in sections 4 and 5.

10.2 Failure witnesses retained by the certificate

The script also checks examples showing that

(i) selecting one signed projective layer changes the actual sum;

(ii) orbit projection and multiplication by the smooth profile do not commute;

(iii) hard windowing destroys exact frequency support;

(iv) residual labels cannot be aggregated before Hilbert projection;

(v) orbit projection is not the same as post-synthesis shift smoothing;

(vi) deleting the progression selector loses two Fourier branches;

(vii) deleting the high profile tail changes the actual coefficient; and

(viii) Hilbert contraction does not imply coordinatewise atomic domination.

10.3 Analytic claim boundary

The following are unconditional theorems of the text:

1. the orthogonal comb projector, its commutation rules, and the exact trace (37);
2. exact projected vanishing for a band-limited progression profile;
3. canonical low/high splitting of the complete actual coefficient and the leakage estimate (78) for the projected frozen family built from the literal physical h_0-coefficient field;
4. the endpoint sector (100);
5. qualitative injectivity on finite orbit windows and the sharp absence of a uniform retained-mass bound; and
6. the exact grouping norm in (88).

The certificate does not test an asymptotic Möbius cancellation statement. Neither the certificate nor the paper proves control of the high-frequency complement, atomic normalization at h_0, sparse alias unfolding, uniform local-shift provenance, a parity breach, the Hardy–Littlewood prime-pair asymptotic, or the twin-prime conjecture.

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